Ranjot Sandhu

Software Engineer | AI/ML Engineer | Full-Stack Developer

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Github: https://github.com/HydraIsProgramming | Portfolio: https://ranjotsandhu.com/

Seeking office, hybrid, and remote roles in the US

SUMMARY

Computer Science graduate with a minor in UX Design and hands-on experience across software engineering, AI/ML, and site reliability. Built machine learning models, automated monitoring workflows, observability dashboards, and user-facing applications through roles at Tata Consultancy Services, Reality AI Lab, and Wilfrid Laurier University. Leverages a UX foundation to make AI accessible and usable for humans, bridging advanced technical capabilities with intuitive, human-centered design. Strong foundation in Python, Java, JavaScript, testing, and scalable system design.

EXPERIENCE

Software Engineer Intern | Tata Consultancy Services (Client: TD Bank) | June 2025 - August 2025

- Developed automated monitoring scripts and SRE tooling that reduced manual incident detection time by 50%+, improving operational response for TD Bank infrastructure.
- Built and maintained Datadog dashboards to monitor production service health and support faster triage across shared banking platforms.
- Collaborated with software and infrastructure teams in an agile environment to deliver reliability improvements for production systems.
- Produced technical documentation, sprint updates, and operational handoff material to support maintainability and stakeholder visibility.

Research Project Developer | Wilfrid Laurier University | September 2025 - Present

- Designed and implemented a predictive machine learning model for a real-world research problem, achieving about 85% accuracy through feature engineering, hyperparameter tuning, and iterative evaluation.
- Built an arm-model research system to identify behavioral patterns in institutional datasets and generate actionable insights for the research team.
- Developed data preprocessing and exploratory analysis pipelines using Python, Pandas, and NumPy to clean, validate, and structure training data.
- Communicated findings through technical documentation and faculty presentations, translating model outputs into research-ready conclusions.

Software Engineer Intern | Reality AI Lab | February 2025 - May 2025

- Developed AI training pipeline components using Python, TensorFlow, PyTorch, Pandas, and Scikit-learn to support model training, evaluation, and validation workflows.
- Performed QA on AI-integrated features using structured test cases, output validation, bug tracking, and regression testing to improve model integration reliability.
- Designed and iterated on product wireframes and UI components in Figma, helping connect AI functionality to clearer user-facing workflows.
- Worked across product and engineering tasks to improve training workflow quality, model output consistency, and release readiness.

SKILLS

Languages
Python, Java, JavaScript, SQL, C, Swift

Frameworks / Libraries
TensorFlow, PyTorch, Scikit-learn, React, Next.js, Node.js, Pandas, NumPy, Matplotlib, SwiftUI

Cloud / DevOps
Datadog, REST APIs, Monitoring and Observability, Automation Scripting, SRE tooling, Supabase, PostgreSQL

Tools
Postman, Fiddler, JMeter, Figma, Git, DataDog, Xcode

EDUCATION

Wilfrid Laurier University - Waterloo, Ontario, Canada
Bachelor of Computer Science, Minor in UX Design
Graduated: August 2026

ADDITIONAL EXPERIENCE

Student Proctor and Teaching Assistant | Wilfrid Laurier University | January 2024 - August 2026
- Supported exam setup, student check-in, and technical troubleshooting for university assessments.
- Coordinated with faculty on exam-day logistics, classroom readiness, and technical requirements.
- Helped create instructional materials including lecture slides and course content.
- Assisted with grading assignments and exams in accordance with course guidelines.

PROJECTS

Bookshelf | Swift, SwiftUI, Supabase, PostgreSQL
/HydraIsProgramming/Bookshelf
- Built a multi-tenant iOS app for retail clerks to track scratch-off lottery ticket inventory and manage end-of-shift accounting across multiple store locations.
- Designed role-based access control (owner vs. clerk) enforced via PostgreSQL Row Level Security across 16 tables, with Supabase handling auth and data — no custom app server.
- Engineered a multi-path barcode scanning system supporting BLE (Socket Mobile SDK), HID-keyboard Bluetooth scanners, and AVFoundation camera scanning with Vision framework OCR fallback.
- Built shift lifecycle RPCs (start, end, mid-shift book swap, force-close) with automated sales reconciliation and per-book inventory state tracking.

Arm Model | Python, R, NumPy, Tkinter, Matplotlib
/HydraIsProgramming/ARM-Model
- Trained a reinforcement learning agent using PPO and SAC to control a robotic arm simulation for daily-living tasks.
- Designed reward functions and policy logic that achieved 90%+ policy confidence against defined task benchmarks.
- Built as a foundational system for future healthcare exoskeleton applications.

EV Infrastructure AI Planning Tool | Python, Scikit-learn, Pandas, Matplotlib
- Built a forecasting tool using historical data to predict Level 2 EV charger demand for residential buildings over a 5-10 year period.
- Validated predictions against regional EV adoption trends and held-out test data.
- Produced planning outputs useful for infrastructure and cost forecasting.

AI Stock Prediction Bot | Python, TensorFlow, PyTorch, SQL
/HydraIsProgramming/Stock_Trading_Bot-AI
- Developed a backtesting framework using historical trade data to evaluate strategies across equities and commodities.
- Generated reports including P&L;, opportunity cost, and alpha.
- Benchmarked model performance against market indices.

3D Portfolio Website | JavaScript, React
https://ranjot-sandhu.netlify.app/ | /HydraIsProgramming/3D-Portfolio-Website
- Designed and built an interactive 3D portfolio website to showcase projects, skills, and experience.

Client-Server Connection Project | Python
/HydraIsProgramming/Client-Server-Assignment
- Implemented a socket-based client-server system supporting bidirectional messaging and file transfer through the command line.

Car Rental Service UI | Java
/HydraIsProgramming/Car-Rental-Interface
- Designed a booking application based on time, distance, and user-defined inputs, exploring optimization ideas for rental and ride-hailing workflows.